

Seeing what's missing

Graphical and computational evaluation of imputation methodology

[TODO: create cover, add boilerplate stuff]

Acknowledgements

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Samenvatting

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Abbreviations and symbols

n	number of units in a sample or rows in a dataset, indexed $i = 1, \dots, n$
p	number of variables in a sample or columns in a dataset, indexed $j = 1, \dots, p$
Y	partially observed sample data (sized $n \times p$), also called the (hypothetically) complete data
R	response indicator matrix (sized $n \times p$), which stores the locations of the observed ($R_{ij} = 1$) and missing ($R_{ij} = 0$) parts of Y
Y_{obs}	observed data (elements of Y_{ij} where $R_{ij} = 1$), also called the incomplete data
Y_{mis}	missing data (elements of Y_{ij} where $R_{ij} = 0$)
Q	quantity of scientific interest; the scientific estimand we would like to estimate if we had complete data
$P(Q \dots)$	complete-data model ; the analysis model we would like to fit if we had complete data, also called the ‘model of scientific interest’
$P(R \dots)$	missing data model ; the model that relates Y to R , which is interpreted as the ‘probability to be missing’; also called the ‘response model’
X	fully observed sample data for covariate(s) of Y , used in missing data models
ψ	parameter(s) of the missing data model that represent any cause of the missingness unrelated to X and Y
MCAR	‘Missing Completely At Random’; missing data mechanism with missing data model $P(R \psi)$, also called ‘not data dependent’
MAR	‘Missing At Random’; missing data mechanism with missing data model $P(R Y_{\text{obs}}, X, \psi)$, also called ‘seen data dependent’
MNAR	‘Missing Not At Random’; missing data mechanism with missing data model $P(R Y_{\text{obs}}, Y_{\text{mis}}, X, \psi)$, also called ‘unseen data dependent’
LWD	list-wise deletion
CCA	complete case analysis
MI	multiple imputation
JM	joint modeling; imputation technique
FCS	fully conditional specification; imputation technique
SMC-FCS	substantive model compatible fully conditional specification; imputation technique
MICE	multivariate imputation by chained equations algorithm
mice	multivariate imputation by chained equations software
m	number of imputations, indexed $\ell = 1, \dots, m$
$P(Y, X, R)$	joint distribution of the incomplete data, fully observed covariate(s), and the missingness

$P(Y_{\text{mis}} \dots)$	joint model, where $P(Y_{\text{mis}} Y_{\text{obs}}, R)$ denotes the posterior distribution of the missing data conditional on the observed data and the missingness
Y_j	univariate subset of Y (sized $n \times 1$); an incomplete variable
Y_{-j}	all data in Y except for column j (sized $n \times (p - 1)$)
$P(Y_{\text{mis},j} \dots)$	imputation model for column j , where $P(Y_{\text{mis},j} Y_{\text{obs},j}, Y_{-j}, R)$ would indicate all other variables are imputation model predictors
$\dot{Y}_\ell$	imputed data in imputation number ℓ (of m total)
Y_ℓ	completed data in imputation number ℓ , consisting of $\{Y_{\text{obs}}, \dot{Y}_\ell\}$
$\hat{Q}_\ell$	estimate of Q calculated from completed data in imputation number ℓ
$\bar{Q}$	pooled estimate based on m imputations
T	total variance of $Q - \bar{Q}$
U	within-imputation variance due to sampling
B	between-imputation variance due to nonresponse
SE	standard error
CI	confidence interval
γ	fraction of information missing due to nonresponse, also called ‘fraction of missing information’ (FMI)
PMM	predictive mean matching; imputation method
V&V	verification and validation of (software) systems, where verification refers to the process of evaluating a system’s output, and validation assesses its fit for purpose.

Introduction

TL;DR (or: the quality of the solution)

This dissertation is about algorithms and the quality of their output. I investigate how one specific type of data-generating algorithms can be evaluated for use in statistical inferences. The type of algorithms in question are imputation algorithms: algorithms with the purpose of generating plausible data values, based on a model of some observed information. This type of algorithm is popular for solving missing data problems.

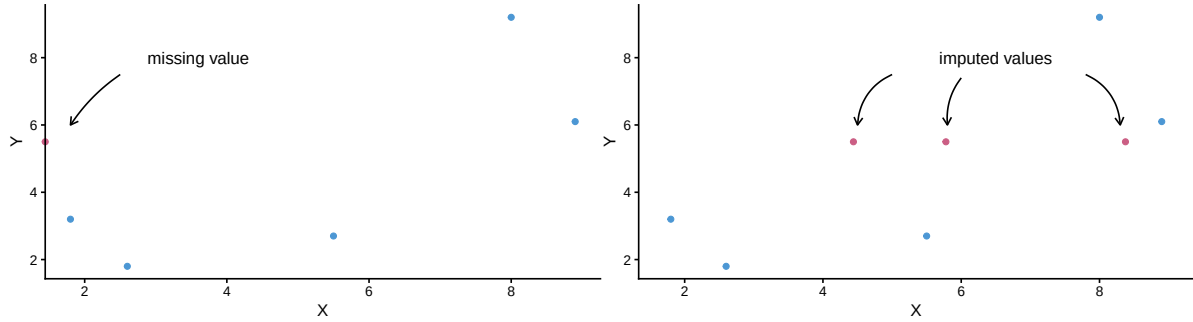
Missing data problems are timeless, but technological advances have made treating missingness easier than ever. A single line of code can seem to solve your missing data problem nowadays. But is that solution really what you expect? That is what my dissertation is about. I want to contribute methodology that facilitates drawing valid inferences from incomplete data.

Missing data are ubiquitous and cannot be ignored without risking bias. Missing data occur often in scientific disciplines with human subjects, such as the social sciences and biomedical research. A participant in a study might, for example, drop out of a longitudinal trial or refuse to answer a sensitive question, resulting in gaps in the research data. These gaps complicate data analysis and may lead to biased and invalid study results. Imputation algorithms are designed to ‘impute’ (i.e., fill in) missing values in incomplete datasets. Imputation is a popular and flexible procedure that separates the missing data problem from the analysis of scientific interest.

Imputation allows data analysts to first solve for the missingness, to then be able to analyze the completed data as if they were completely observed. If done several times in parallel, imputation can yield estimates that are both unbiased as well as confidence valid. The practice of drawing inference by replacing missing values with several synthetic substitutes is called ‘multiple imputation’ (MI). Figure 1 provides an example of multiple imputation, where a missing value is replaced several times with plausible substitutes. The variability between imputations will reflect the uncertainty due to non-response in the final estimate.

Although imputation is widely supported by software and many implementations of the method have convenient defaults, generating good imputations is not a trivial task. How can we know when to trust the output of imputation algorithms? There are three approaches. The ideal route is to study the formal statistical properties of the method via analytical derivations. Unfortunately, this is often impossible for imputation algorithms, many of which do not have formal proofs. Instead, the de facto standard is to estimate their empirical performance via simulation studies. These studies, however, are not standardized and are therefore hard to compare. Moreover, simulation study results might not always translate to applied use. Thirdly, for these real-world applications there is no systematic framework for evaluating the practical performance of imputation algorithms, such as diagnostics for algorithmic convergence. Ergo, evaluating whether imputations should be trusted remains challenging.

Figure 1: Multiple imputation of a missing value



In this dissertation, I set out to investigate what is needed to develop, apply, and evaluate data-generating algorithms. I thereby focus on the imputation procedure ‘MICE’ as implemented in the R package *mice*.

Why? Because current evaluation practices (simulation design, convergence diagnostics, visual diagnostics) are fragmented and insufficient.

How? By bridging inferential evaluation and data-/algorithm-level diagnostics for imputation methods.

Why this dissertation? (or: statistical inference is just a tool)

My academic tagline has long been “statistician, interdisciplinary, open scientist”. And while there’s no doubt I am an interdisciplinary and an open scientist, I am no longer keen on calling myself a statistician necessarily. I am more inclined to say I am a methodologist rather than a statistician. As a methodologist, I am interested in how science is done, and how it can be done better. Especially in the social and biomedical sciences, research should not be purely quantitative. Statistics is just *one* way of doing science.

Don’t get me wrong, statistics is still a favorite of mine. I love stats (I even have a mug that says so). Statistics is the science of uncertainty and extracting information from data (Hand, 2025). Extracting information from data and turning them into insights can feel magical. Statistical inference allows us to distinguish between real-world effects and random noise (Rosenberg, 2018). I am most a fan of the uncertainty aspect.

I am motivated to contribute to trust in science by making it easier to reflect uncertainty in data analysis efforts, inferences, and graphs. In this dissertation, I hope to achieve this for missing data methodology.

But first, some background and notation

Statistical analyses are a means to infer conclusions about a population after observing only a subset of this population: a sample. If the sample is representative, sample estimates will reflect the population characteristics plus some uncertainty surrounding it. Statisticians call this uncertainty the sampling error, due to observing only a subset of the population, which is often expressed as part of the standard error (SE) surrounding the estimates. Standard errors then get translated into decision tools for inference such as p -values and confidence intervals (CIs), which help researchers distinguish between signal and noise. So, statistics is a means of estimating effects within a sample and inferring back to the population with some expression of uncertainty.

Uncertainty in statistical inferences can stem from many sources. In the Total Survey Error framework (Deming, 1944), other sources of uncertainty are described such as non-response error. Non-response error occurs when participants do not (fully) respond to data collection efforts. We speak of unit non-response for intended study participants who do not provide any data at all (for example because they could not be reached, or refused to participate). Item non-response refers to patterns of missingness where some—but not all—data are missing for a participant (for example because they skipped a survey question). Item non-response is the focus of this dissertation: the way we handle incomplete observations in partially observed data and how that affects our inferences.

How we treat non-response is reflected in the uncertainty of our inferences. Unit non-response is often treated using weighting, which may lead to weighting error Alsalti et al. (2026). It is not typically part of the TSE framework, but item non-response treatment can also produce error, such as ‘imputation error’ (Federal Committee on Statistical Methodology, 2001).

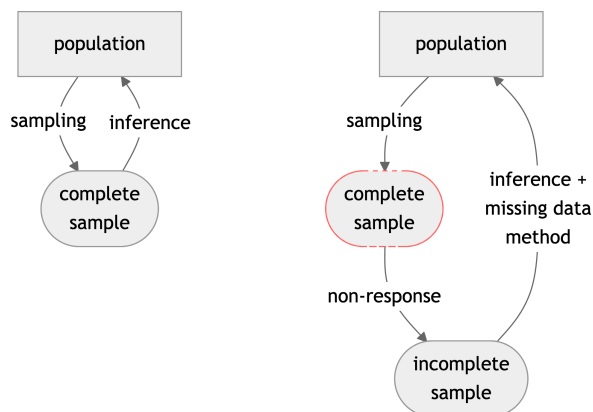
Non-response bias can be minimized by preventing missingness altogether: the best way to treat missing data is by not having any (truism attributed to Gertrude Mary Cox). Some precautionary measures... For example by motivating participants to ... Whenever missing data cannot be prevented, the only way forward is to treat the missing data problem. Missing values force data analysts to treat the missingness in some way.

Why are missing data problematic?

Let’s use an empirical example. Generally speaking, statistical inference works really well. For example, when sociologists wanted to study trust in the government during the COVID-19 pandemic, it was not feasible to survey all Dutch citizens. Instead, they used a representative sample. Some thousands of people were asked to fill in a questionnaire about their trust in the government over the course of the pandemic. By repeating the questionnaire over time, the researchers wanted to gain insight into the development of (dis)trust among different subgroups in Dutch society. The estimates from this sample would have been used to infer back to the population and inform policy, if it had not been for missing data...

Missing data are a nuisance because it is impossible to do statistical inferences with incomplete data. When there is just one missing value in a dataset, statistical inference cannot be performed. With incomplete data, even simple statistics such as the mean of variables are mathematically undefined. To obtain results from your incomplete dataset, you will always employ a missing data strategy—whether deliberately or not.

Figure 2: Inference with incomplete data



Note. Schematic view of statistical inference in the presence of non-response. Left: no missing data in the sample from the population (i.e., ideal circumstances/business as usual). Right: incomplete sample due to non-response, so no complete sample could be obtained. What types of missing data problems are we considering? We are interested in data with incomplete cases and/or variables, or 'item non-response'. Incomplete samples where some—but not all—entries are missing per row or per column. Entirely missing rows would be 'unit non-response'; entirely missing columns would be unobserved variables.

To get an estimate from incomplete data, some kind of missing data method has to be employed. Even without specifying a missing data method explicitly, there will be handling of the missing data. The default strategy in most statistical software packages is to ignore all cases that have missing values (van Buuren, 2018). This is referred to as 'list-wise deletion' (LWD). Ignoring missingness, however, lowers statistical power and may yield invalid inferences (e.g., biased estimates, spurious significant results; (van Buuren, 2018)). List-wise deletion is ill-advised under most conditions (see van Buuren 2018 for exceptions). List-wise deletion results in bias because it assumes that the observed part of the data is equivalent to the missing part of the data, and this is not usually the case.

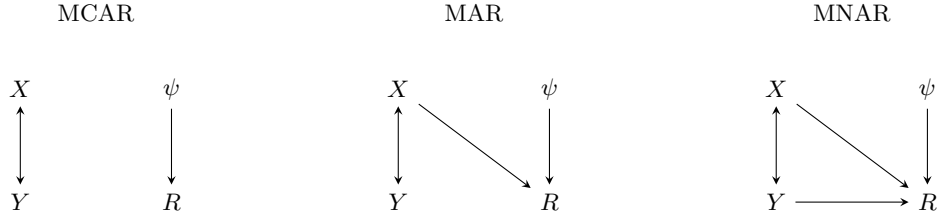
Why are the missing and observed parts not equivalent? (or: the missing data mechanisms)

List-wise deletion (complete-case analysis) assumes that the observed part of the data is representative of the missing part of the data, which seldom holds in practice. In practice, the missing part of the data is often missing due to some non-trivial reason, other than pure chance

(or ‘bad luck’). If there is a non-chance cause for the missingness across observations, the observed part of the data is not representative of the missing part of the data. This relationship can be formulated in terms of a missing data model. Missing data models capture the causal relation between the data (Y) and the associated missingness indicator (R). Missing data models thereby reflect the ‘probability to be missing’ for any given observation. Missing data models are used to describe and classify the causes of the missingness, which determines how the missingness should be handled to arrive at unbiased and confidence-valid results.

There are three missing data mechanisms described by missing data models. If there is no data-related cause cause of the missingness, the missingness is said to be ‘completely at random’ (MCAR), or ‘not data dependent’ (respective terminology by Rubin, 1976; and Hand, 2020). If there is an association between the missingness and the observed part of the data, the missingness is ‘at random’ (MAR; conditional on the observed data) or ‘seen data dependent’. If the cause of the missingness is missing too, the missingness is ‘not at random’ (MNAR), or ‘unseen data dependent’. Figure 3 provides a graphical representation of these three mechanisms, where the missingness (R) in incomplete data Y either depends on Y itself (i.e., MNAR), on observed covariate(s) X (i.e., MAR), or none of these (i.e., MCAR).

Figure 3: Missing data mechanisms



Note. Incomplete data Y and fully observed covariate X covary. Under missing completely at random missingness (MCAR), the missingness indicator R is only affected by ψ , causes of the missingness unrelated to X and Y . Under MAR, the missingness also depends on observed information (X), whereas MNAR entails missingness that is dependent on the missing data itself (Y). Figure adapted from Schafer & Graham.

Modeling the non-response can aid the analysis of incomplete data. Together with complete-data models (i.e., the model of scientific interest), data models inform data analysts how the missingness might affect the scientific estimates. For example, in that study about trust in the government during the COVID-19 pandemic, it was not random who had missing data on the question about trust in the government. People with lower trust in institutions were also less likely to respond to the questionnaire. This resulted in higher missingness rates across participants with lower trust in the government. Using only the observed data to infer to the population would then leads to biased estimates (over-estimation) of trust in the government. Methods that implicitly assume MCAR (like list-wise deletion) are therefore likely to cause bias.

In human subjects research, pure MCAR missingness is unlikely. Only truly random missingness, independent of any individual characteristics, would amount to missing completely at random

missingness mechanism. This could happen, for example, due to a glitch in the service software, affecting all participant equally likely. In our case, however, there were clear signs of non-MCAR missingness. If we can model the missingness probability based on known data, such as previous observations or demographic data, we have ‘Missing At Random’. Then, conditional on this information, observed and missing values have the same probability to be missing. Under MAR, we can model the missingness probability using known variables and obtain valid inference if this model is adequate. This is in contrast to the ‘unseen data dependent’ missingness mechanism or ‘missing not at random’. If the missingness depended the unobserved values themselves, we speak of ‘Missing Not At Random’. For example, when people who distrust the government are less likely to respond to the trust item, and we have no way of correcting for this. This kind of missingness cannot be resolved from the observed data alone and requires assumptions or external information. In practice, it is impossible to know from the observed part of the data alone, so we have to make assumptions. Differences between the observed part of the data and the missing part can be accommodated by treating the missing data problem.

How to treat the missing data problem? (or: ignore, model, fill in – the missing data methods)

This section describes three popular routes for treating missing data. I will not pretend to provide an encompassing overview on treating missing data problems. Others have written much more extensive reference works on the matter, see for example Rubin (1996), Schafer (1997), Allison (2002), Schafer & Graham (2002), Graham (2012), Molenberghs et al. (2014), van Buuren (2018), and Little & Rubin (2019). The three missing data treatments I will describe are deletion-based methods, likelihood-based methods, and imputation.

The first missing data treatments are deletion-based methods, such as list-wise deletion. With list-wise deletion, we only analyze rows in the data without missing values (i.e., complete cases). Deletion-based methods are easy to apply, and sometimes even applied by default in statistical analysis software. If the missingness level is low, deletion-based methods do little harm, and under some conditions they might even be the best missing data method to apply (see e.g., van Buuren, 2018 § 2.7; Carpenter & Smuk, 2021, § 3.1). Deletion-based methods, however, do not generally accommodate differences between the observed and missing parts of the data, making them mostly applicable to MCAR. Under most kinds of missingness, deletion-based methods will therefore risk biasing the statistical inferences. Moreover, these methods discard incomplete observations, resulting in lower power and potential resource waste compared to non-deletion-based methods. In short, deletion-based methods do not treat the missing data as much as ignore them, and one should be careful of bias when applying them.

Secondly, there are methods that incorporate the missingness into the model of scientific interest. Modeling methods do accommodate MAR, but are often quite complex. These missing data methods jointly specify the complete-data model and missing data model per analysis. For example, Bayesian methods can incorporate the missingness into the analysis model. But you have to specify the model for each analysis separately, and there are no null hypothesis

significance tests which is a problem when working with applied researchers. Another option to incorporate the missingness in the analysis model is expectation maximization. But this yields many parameters and might not be available for a complex multilevel model like the COVID-19 data. Both of these techniques are tied to a specific analysis model, which might not be suitable for contemporary practitioners who run several models in a data science workflow.

Thirdly, there are imputation methods. Imputation methods do not have these drawbacks of being computationally advanced and analysis-model specific. Imputation splits the missing data problem from the analysis of scientific interest. Imputation methods first solve for the missingness, to then analyze the completed data ‘as if’ they were fully observed. This allows the use of familiar complete-data methods instead of bespoke incomplete-data models. Imputation is useful when different people are responsible for different parts of a project (e.g., a missing data methodologist vs. domain expert). And when many different analyses are planned on the same data, it is possible to run several analyses without having to incorporate the missingness into the scientific model each time. This makes imputation better suited for contemporary data science workflows than missing data methods that incorporate the missingness in the substantive model. Compared to missing data methods that incorporate the missing data like Bayesian methods or likelihood-based approaches (e.g., EM), imputation is more flexible and often less complex for non-experts. Compared to list-wise deletion, imputation methods offer the ability to accommodate MAR mechanisms, and they reduce research waste by retaining incomplete cases. Imputation thus has the ability to produce unbiased estimates. But are they properly uncertain (i.e., confidence-valid) too? That’s where multiple imputation (MI) comes in!

Why multiply impute? (or: several ‘wrongs’ *can* make a ‘right’)

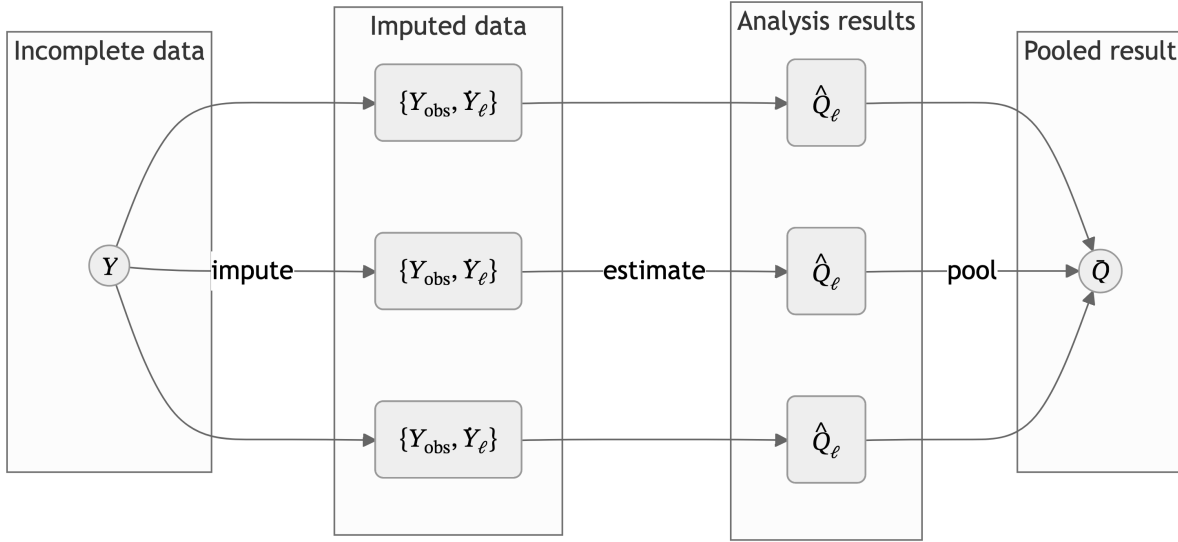
All techniques that rely on a single replacement value risk under-representing uncertainty in the inferences, because the uncertainty due to the non-response is not taken into account. The uncertainty due to non-response can be captured using the multiple imputation framework. Conceptualized by Rubin (1976), MI posits that repeated stochastic draws per per missing datapoint create a distribution of plausible values: the posterior distribution of the missing data conditional on the observed data and the missingness, $P(Y_{\text{mis}}|Y_{\text{obs}}, R)$. This posterior distribution reflects the uncertainty due to non-response. By drawing imputations from the posterior distribution of the missing data, we obtain completed versions of the data that *might have been*.

MI is the practice of generating several completed datasets by replacing each missing value with a synthetic substitute multiple times. The completed datasets can be analyzed as if the data were complete to begin with. On each completed dataset, we obtain estimates of the analysis of scientific interest. These estimates per imputation are then combined to obtain one pooled estimate. The variance of the estimate incorporates not just the sampling variance (within each imputation) but also added variance due to non-response (between imputation

variance and simulation error). The variability between imputations reflects uncertainty due to the missingness.

The result is a probability distribution for each missing value, from which we subsequently draw several values as imputations. If we would use infinitely many values as imputations, we would obtain the entire probability distribution. In practice, however, we use a limited number of imputations. With that, we approximate the probability distributions of missing values. The more imputations are used, the better we can represent the uncertainty in the data due to missingness. This is expressed in the ‘simulation error’.

Figure 4: Multiple imputation scheme ($m = 3$)



Note. Read left to right. Incomplete data Y is partially observed, $Y = \{Y_{\text{obs}}, Y_{\text{mis}}\}$. The missing data Y_{mis} is imputed m times to obtain m completed datasets $\{Y_{\text{obs}}, \hat{Y}_\ell\}$, where $\ell = 1, 2, \dots, m$. On each imputation, the quantity of scientific interest Q is estimated, yielding m analysis results $\hat{Q}_\ell$. These results are pooled across imputations to obtain a single pooled result $\bar{Q}$.

Multiple imputation means that each missing value is filled in several times, with slightly different model parameters to represent model uncertainty. The result of MI is that the uncertainty in the data due to missingness is preserved in the inference. Correcting for missingness happens at two levels: 1) modeling the relation between missing cases and covariates; 2) adding the appropriate variance to the models to represent the uncertainty due to missingness. Under the right conditions, MI yields valid, unbiased estimates (Schafer, 1997). MI enables researchers with missing data problems to correct for differences due to missingness and maintain the appropriate level of uncertainty. The aim is to generate imputations to fill in the missing part of the data. How do we model the missing part of the data based on the observed part of the data?

How to generate imputations? (or: univariate, joint, chained – the imputation flavors)

We model the distribution of this missing part based on what is observed, using the observed part of the data to fix the missing part of the data. Which types of model or models do we use?

The simplest model is no model at all. With hot deck imputation, we use random other person's observation to impute or fill in the values for people with a missing data point. The problem is that this disrupts the multivariate distribution of the data, so we get biased estimates. Another simple technique is to impute a univariate statistic such as the mean. Unfortunately, imputing the mean will bias your scientific estimates even worse than list-wise deletion would, because mean imputation severely biases any multivariate statistics downward, and there is no correction for the missing at random missingness mechanisms. In general, all univariate imputation methods fail to retain multivariate associations in the data, and therefore cannot account for MAR mechanisms. So, instead of imputing variables univariately, imputations should retain the multivariate structure of the data.

Capturing the multivariate structure of incomplete data is historically done in two steps: a modeling step and an imputation step. In the modeling step, a multivariate distribution of the full dataset would be defined, after which imputations could be drawn from the joint model. In a joint model, one specifies $P(Y, X, R)$: the distribution of the incomplete data, complete covariates, and missingness indicator. Joint modeling imputation was originally developed for known distributions, such as multivariate normal data. With increasing data complexity and dataset size, specifying a joint model might not be feasible. The problem with joint modeling is that there might not exist a multivariate probability distribution, for example, with factor variables involved. Joint modeling is a direct and inflexible way to arrive at a joint distribution. Good thing we do not need this joint model explicitly.

Missing observations can be filled in using one or more imputation models instead. While Rubin (1987, pp. 160–166) “distinguished three tasks for creating imputations under an explicit model: the *modeling* task the *imputation* task and the *estimation* task.” (van Buuren, 2007, p. 221; in Molenberghs et al., 2014), FCS does not require these explicit tasks. In the FCS framework (van Buuren et al., 2006), we only need an imputation model for each incomplete variable. Imputation models consist of the functional form of the model (e.g., stochastic regression) and imputation model predictors (i.e., other variables in the data). FCS is substantive model agnostic, and does not assume any specific joint distribution. Instead, FCS only requires an *imputation model* to describe how synthetic values may be generated, without explicit specification of the joint model. This makes FCS flexible. Just specify an imputation model per variable.

How to define imputation models?

We define an imputation model per incomplete variable. And we need two things for this: 1) We need to specify the functional form of the model. For example a semi parametric method such as predictive mean matching for Likert scale items such as trust in the government. And 2) we need to choose imputation model predictors. For example, variables with a high linear association with this incomplete variable, or variables that are related to the missingness indicator of this incomplete variable (such as demographic data).

Some things to keep in mind: With FCS, the quality of the imputation model can only be determined/assessed based on the complete data model. But what if that model is not (yet) available? Then take the most complex possible model and check that, because if you would take the simplest model, then that would implicitly mean you are constraining parameters (forcing $b=0$), systematically suppressing relations. Don't forget to add the effect of interest in your model of scientific interest. Otherwise, you are constraining the relation between the variables to be zero. If there is a specific data analysis model for the effect of scientific interest that you already have in mind, consider substantive model compatible fully conditional specification. SMC-FCS is optimized for known analysis model models: we assume a complete data model and derive the imputations from there. But what if the analysis model is unknown? Well, Fully conditional specification is the substantive model agnostic. So we build an imputation model based on the most complex analysis model that we might want to test.

FCS implicitly relies on the joint distribution. The nice thing about joint modeling is that convergence of the model is guaranteed, whereas this is not the case with fully conditional specification. "Imputation models bypass the need to specify $P(Y, X, R)$, though their use creates new responsibilities for substantiating its correctness for a given statistical analysis." (van Buuren, 2007, p. 222). The question becomes: how do we choose an appropriate imputation model? And how do we know whether we have chosen an appropriate imputation model?

FCS has no deductive or analytical proof, so we have to rely on evidence from simulation studies. It has been shown that FCS works under many conditions. But how to know it works in practice? We need assumptions. Many modeling assumptions inherent to a model-based technique like MI cannot be tested without the missing data itself. So, like assumptions for GLM, we look at the data to evaluate the appropriateness of our chosen model. Similarly, we need to evaluate the choice of imputation models for signs of assumption violations. Some assumption checks are visual inspection, some are quantitative, some need external info.

How to evaluate imputation models?

There are three main routes to evaluate imputation models. One can evaluate:

- the formal statistical properties of the method via analytical derivations;
- the empirical performance (estimates of the method's attributes) via simulation studies;
- and

- the operational utility (practical usefulness) in real data.

The first and preferred option is to obtain the formal statistical properties of the imputation method using analytic derivations. Often this is impossible or unfeasible with fully conditional specification, when no known joint model exists. Without this, we cannot definitively conclude that an imputation method is an unbiased estimator. Instead, we estimate the empirical performance of these methods using simulation studies. This is the default evaluation in the missing data methodology field.

Simulation studies are the standard, but not standardized. If we have a specific analysis model of interest, we can calculate performance metrics such as bias and confidence interval coverage to conclude whether a method is unbiased and confidence valid for a certain estimand. This requires a specific analysis model, which might not be the case. And it is impossible to cover all the potential use cases. Another problem is that this is not standardized, so it's difficult to compare different simulation studies. We could also calculate the imputation accuracy by comparing the imputations against the complete data, but this is ill-advised.

The third option is the operational utility. This is case-by-case evaluation in applied use with real data. We check the imputation model itself, for example to see whether the algorithm has converged. We also inspect the output and we might run sensitivity analyses. This is not systematic at all currently. That is a problem. Users of the method then do not know whether they have correctly applied the method.

In this dissertation, I focus on the second and third routes. We will borrow some terms from related fields. I will connect it to concepts from systems and software engineering, and the information retrieval sub-field of computer science.

Computational evaluation, verification and validation [TODO: edit]

In computer science, software is a system. “A system is a purposeful collection of interrelated components of different kinds (e.g.,”hardware, software, and human elements” p. 552] that work together to deliver a set of services to the system owner and its users.” (Sommerville, 2015, p. 556) Systems engineering is dedicated to the study and development of systems “to support human activities” (Sommerville, 2015, p. 552). “Systems that include software fall into two categories: 1. Technical computer-based systems [and] 2 Sociotechnical systems”. (Sommerville, 2015, p. 552) “For example, this book was created through a sociotechnical publishing system that includes various processes (creation, editing, layout, etc.) and technical systems (Microsoft Word and Excel, Adobe Illustrator, Indesign, etc.).” (Sommerville, 2015, p. 552) “Sociotechnical systems: include one or more technical systems but, crucially, also people, who understand the purpose of the system, within the system itself. Sociotechnical systems have defined operational processes, and people (the operators) are inherent parts of the system.” (Sommerville, 2015, p. 552). So imputing is a sociotechnical system, not technical system, while mice is technical. “sociotech systems “do not just include software and hardware but also people, processes, and organizational policies” p. 556] That means that mice is a technical

system, whereas the imputation workflow is sociotechnical and depends on other components. “The successful functioning of each system component depends on the functioning of other components” (Sommerville, 2015, p. 556). “Systems have properties that only become apparent when their components are integrated and operate together.” For imputation this means that there should be ‘interoperability’ with other systems, such as ggplot for viz, and stats packages for analyses.

Why is validation needed? “Sociotechnical systems are complex systems, which means that it is practically impossible to have a complete understanding, in advance, of their behavior. This complexity leads to three important characteristics of sociotechnical systems: 1. They have emergent properties that are properties of the system as a whole, rather than associated with individual parts of the system. Emergent properties depend on both the system components and the relationships between them. Some of these relationships only come into existence when the system is integrated from its components, so the emergent properties can only be evaluated at that time. Security and dependability are examples of important emergent system properties. 2. They are nondeterministic, so that when presented with a specific input, they may not always produce the same output. The system’s behavior depends on the human operators, and people do not always react in the same way. Furthermore, use of the system may create new relationships between the system components and hence change its emergent behavior. 3. The system’s success criteria are subjective rather than objective. The extent to which the system supports organizational objectives does not just depend on the system itself. It also depends on the stability of these objectives, the relationships” (Sommerville, 2015, p. 558)

In systems engineering, the system life cycle framework is used to build, develop and evaluate systems. There are standards for the evaluation of a system: verification & validation. V&V is needed when the ‘high road’ of analytical derivations and proofs are impossible or unfeasible. Instead, computer scientists use computational evaluation to assess systems.

Verification asks ‘was the system built right?’, whereas validation asks ‘was the right system built?’. So with verification, we can check the quality of a system by plugging in an input with a known output. Do we get the correct output in return? This is what we do with simulation studies and with unit tests in software. We want to check whether the system or the method in this case conforms to the requirements that we set (e.g. accuracy). Verification is about conformity to specifications, usually before or independent of real-world use.

Then validation, on the other hand, asks whether the system is suitable for operation by the actual users in a specific intended use or application. Is a system able to accomplish its intended use? We can, for example, use integration tests. Validation = use of use cases and usability testing to evaluate whether users have all they need to use the system. In stats, it’s proving that a statistical model or analytical procedure behaves as intended under real-world operating conditions. Validation is often socio-technical: it combines technical inspection with **human judgment** and domain knowledge. Validation test = “test to determine whether an implemented system fulfils [sic.] its specified requirements” [ISO/IEC 2382:2015, Information technology — Vocabulary]

How do methodologists know whether an imputation method works? Because we have empirical estimates from simulation studies (and, if available, formal properties from analytical derivations). How do imputers know whether their imputation models worked? Our current answer is well, they don't know for sure. And that's what this dissertation is all about. We need computational evaluation and diagnostic visualization.

Computational evaluation encompasses a range of systematic, computer-based evaluation tools such as numerical experiments or simulation studies. Computational evaluation = use of metrics, tests, and simulations to evaluate properties of the system with minimal human judgment.

Diagnostic visualization is the use of graphs to communicate system behavior to stakeholders. It's 'diagnostic' because the purpose is to identify patterns and problems instead of descriptive/summaries. Diagnostic visualization/visual inspection helps stakeholders understand the system's behavior and judge whether it's acceptable for their needs.

Evaluating imputations

Life cycle of imputation methods: method is theorized, method is implemented in software, method is evaluated in simulation study, method is put to scrutiny by scientific community, method is applied in practice, method is evaluated in real-world data.

- Step 1: verification of statistical properties via formal analysis (e.g. proofs) and/or simulation
- Step 2: validation of operational utility via graphical and computational evaluation

Step 1

Gold: analytical properties. Deductive truths. But not feasible with FCS. So, implementation needed, which does require V&V.

Silver: simulation studies. Statistical evaluation is the default to evaluate the inferential validity. This is verification. Assessing whether the system yields correct (known) output when a specific (known) input is supplied. Examples are unit tests for software(?) and performance assessment of methods via simulation. Verification via simulation is done, but not standardized. Verification via testing is mostly done as unit tests, but not as integration tests.

Step 2

Operational utility. This is validation. In practice, you cannot validate whether the imputation model was appropriate, because what you would need to do so is exactly the thing you don't have: the missing part of the incomplete data, while you only have the observed part. The thing we need is not available by definition. Inferential validity cannot be established. Some assumptions are impossible to definitively determine (e.g. MNAR), some we can evaluate by

looking at the data. The setting where we do have the comparative truth (the true but missing values) is simulation studies. Imputers have to rely on simulation study results in order to choose the best imputation method for their incomplete variables. That’s why simulation studies for imputation methodology should be comparable with one another. Then, imputers can make a ‘grounded’ assessment which imputation methods may be applicable, and choose the most appropriate one. Example: NRI search for lower bound, so impute as ‘positive behavior’, but per definition wrong statistical inf, because we intentionally bias in one direction

In simulation studies we can construct the “truth”, but in practice we have to rely on assumptions. Those assumptions are crucial precisely because they cannot be (fully) proven. In practice, the best thing we can do is to rely on a combination of assumptions/theoretical justifications/substantive expertise and indirect evidence of imputation model fit: check the observed versus imputed data distributions, and inspect the imputations for signs of non-convergence. Computational (quantitative) evaluation – metrics, tests, simulations. Graphical (qualitative) evaluation – diagnostic visualization/visual inspection. This dissertation offers insight and tools to do so.

What is computational evaluation? [TODO: edit]

Computational evaluation is originally from information retrieval research (see e.g. “Evaluation Measures (Information Retrieval),” 2025) The practice of computational evaluation combines *effectiveness* (i.e., did the process yield the correct result, e.g., accuracy, precision, recall, etc.), *system quality* (e.g., speed, coverage of all possible results, etc.) and, importantly, *user utility* (e.g., happiness, productivity, etc.) (Manning et al., 2008) The term ‘computational evaluation’ is not specifically defined for the field of statistics, but it could be useful. One definition of computational evaluation is “to determine the quality of solutions attainable” (Dudek, 2004) Translated to statistics, a model may be statistically valid, but not fit for purpose. For example, computations that take too long to converge for use in a production pipeline/real-time use in patient care. Another definition of computational evaluation is “the process of ensuring an algorithmic solution is a good one: that it is fit for purpose” (Curzon et al., 2014). We can use this when a statistical process has different purposes and thus different usability/requirements. For example, calculating uncertainty at the level of a single case, a sample, or inferring to a population. Since the term is not defined for statistics, there is no definition to use for the computational evaluation of imputation methodology. In this dissertation, I want to plant a flag: how can we define ‘computational evaluation’ in the context of imputation methodology? Working definition: *systematic assessment of the algorithmic behavior and data-level outputs of imputation procedures*, focused on: algorithmic stability (convergence, failure modes), sensitivity to modeling choices, plausibility and coherence of imputed values, and usability for applied researchers. These aspects are often already taken into account by imputers, but I try to systematize and operationalize these aspects that are often informal or *ad hoc*.

Computational evaluation is not equal to statistical evaluation. Computational evaluation goes beyond statistical evaluation. Statistical evaluation is important, e.g. to quantify bias

and variance of estimators/procedures, but not the focus of this dissertation. Computational evaluation is something to check *conditional on* acceptable statistical properties, and does not replace statistical evaluation. Dimensions of computational evaluation: e.g. inferential validity, algorithmic behavior, data-level plausibility, user utility, ...

Why should we care about computational evaluation of imputation methodology? [TODO: edit]

What do we care about? bias/coverage/variance, empirical relevance/plausibility, software-engineering nitpicks like speed/convergence/fault rate. Statistical evaluation is not possible in practice, only in simulation studies. Why is statistical evaluation may not possible IRL? Bias and confidence interval coverage are impossible to determine without a comparative truth (although PPC and bootstrapping allows us to compare with respect to observed values). In this dissertation I will use computational evaluation as a suite of tools to investigate the domains of algorithmic stability, data plausibility, and user burden, next to inferential validity.

User utility does not only require statistical validity, but also usability/appropriateness/plausibility of the imputations. Even if we know the process yields correct inferences, we can evaluate the ‘fit’ of the solution.

One question to consider is: do the imputed values lie within the range of the observed values? For statistical validity this is not needed. Imputation theory does not require the imputed values to be plausible. Rubin (1976, 1987) developed the methodology to get correct population-level inferences from incomplete samples, not at the individual case-level within the sample. [TODO: consider adding that since then, data analysis has changed] Bluntly stated: imputation pragmatists/purists don’t care for the imputed values themselves, as long as they result in statistically valid inferences. But end-users (e.g. clinicians) do care for plausibility. For example, imputing negative values for age will yield ‘unrealistic’ imputed values. Sometimes this is bad, such as using PMM for a variable with a sparse region, because PMM with part of sample space with too little donors might lead to bias. Sometimes this is okay, such as with polynomials (ignoring the U in imps *could* still get statistical validity).

What are other limitations of purely inferential metrics (bias, coverage)? Statistical evaluation relies on known analysis models, while FCS is substantive model agnostic. So, supplement it with computational evaluation and visual inspection. These are also possible without analysis model, which makes it more pressing today, because imputations might be used with several analyses. Not knowing the analysis model up front risks non-congeniality

Another problem of unknown applications with potential non-congeniality is non-convergence. Without graphical or computational diagnostic for non-convergence, users might analyze imputed values of non-converged chain. For example, fcs iterations not stable, but estimand okay (because in simulation study we have true value), but what about conv at cell level? Statistical models like MI are optimized to distinguish between effect and noise of population averages—not the individuals that make up your sample.

But also vice versa: when might the plausibility of imputed values mislead a data analyst into believing the imputations are statistically valid? When ‘looking reasonable’ is confused with ‘being consistent with the data-generating process’, such as single imputation (stoch reg), or imputing under MCAR assumption when there is a MAR mechanism, or destroying the joint distribution of the data while preserving the marginals. Another example: psych study with baseline, some of which are missing; baseline, intervention, follow-up → mice would inflate baseline and follow-up if both are incomplete; statistical evaluation will not show this (unbiased, cov ok, etc.); over-imputation or postpred check would show diff distr for obs and imp; circularity should be removed: follow-up should not predict baseline So computational evaluation should never replace statisticeval.

What is the scope of this dissertation? [TODO: edit]

In this dissertation, we will focus on missing data problems with item non-response and MCAR/MAR missingness mechanisms. You might expect me to go into...

- In the category non-response: just item non-response (not unit non-response), and M(C)AR mechanisms (not MNAR). Unit non-response and MNAR missingness mechanisms are outside of the scope of this dissertation.
- In the category imputation algorithms: just FCS in the flavor of MICE (not JOMO or SMC-FCS). The missing data method under consideration is primarily multiple imputation by chained equations (MICE), but some results may generalize to other iterative (FCS) imputation algorithms. We do not cover joint modeling (JOMO) imputation or non-imputation missing data methods. Although MICE is not SMC-FCS, we will restrict the analysis of scientific interest mostly to GLM family of models.
- In the category substantive models: just GLM for statistical inference. The applications are aimed at statistical inference (not prediction or causal inference).

RQ: what tools are required to assess the quality of data-generating algorithms? in the development, evaluation, and application.

What is the status quo? Verification via statistical evaluation (but not standardized), and some validation via computational and graphical evaluation (but not systematic). Is the status quo sufficient? No, statistical evaluation is required, but not sufficient. How Rubin developed the methodology works in theory, but is not enough for contemporary practitioners.

Can we use existing tools or do we need more? and how? No, not all tools are present in the imputer’s toolbox. I contribute to the new status quo (“extending” = generalizable scientific contribution of this dissertation) by unifying how imputations are evaluated across the life cycle. There two phases: statistical properties of the method in simulation (e.g., bias and confidence-validity; statistical evaluation) & practical application (graphical and computational evaluation).

Chapters

1. Evaluation paper: evaluating (new) imputation methods via simulation studies/how we know FCS works
2. Convergence paper: evaluating imputation algorithm quantitatively
3. Software: choosing/evaluating imputation models in practice

Introducing chapter 1: evaluation [TODO: edit]

- **RQ: what to look out for when designing simulation studies for the evaluation of imputation methodology?**/How can computational evaluation criteria be systematically incorporated into simulation studies for imputation methodology?
- published as Oberman & Vink (2024), cited 27x according to Google Scholar (of which 11x CrossRef)
- based on literature review Liu et al. (2023)
- How can we trust simulation studies for imputation methods?
- How can we make them compatible & fair?

REF: Morris ADEMP: Using simulation studies to evaluate statistical methods, pre-registration template, FIMD, Introduction to statistical simulations in health research?, Pitfalls and Potentials in Simulation Studies?

Introducing chapter 2: convergence [TODO: edit]

- **RQ: FSC is iterative, so convergence *should* be a requirement for valid inference, or is it?**/How does non-convergence in iterative imputation affect inferential validity, and how can non-convergence be diagnosed?
- simulation study, published as pre-print (11x cited according to Google Scholar), presented at ICML
- even without convergence, we can get valid inference at sample/population level, but not at individual cell level
- imputation stays a custom job: convergence of imputation is conditional on imputation model
- today, imputation are not generated for 1 analysis prob, but all analyses that may be fitted on the data -> all imputation models should be compatible and congenial -> more 'omvattende' imputation models required

- liefst wil je analysemodel parameters tracken, daarom is computational evaluation zo belangrijk
- Can we quantify the convergence of the MICE algorithm?

REFs: Graphical and numerical diagnostic tools to assess suitability of multiple imputations and imputation models, Diagnostic checking of multiple imputation models, Convergence Properties of a Sequential Regression Multiple Imputation Algorithm, FIMD, Diagnostics for multivariate imputations

Introducing chapter 3: `ggmice` [TODO: edit]

This chapter introduces the software `ggmice`: an R package for building and evaluating imputation models. The `ggmice` package offers tools for the visualization of incomplete data, imputation models, and imputed data.

Indicators for adoption: how is the software already used?

- software published on Zenodo (Oberman, 2022), with open code and bug reporting via GitHub
- peer-review not formal via journal, but via Issues and Pull Requests on GitHub (10x external Issue, 13x external PR)
- impact: citations (5x according to Google Scholar), CRAN downloads (852x/month; 21k total) and conda downloads (100+/month; <https://anaconda.org/channels/conda-forge/packages/r-ggmice/overview>)

Design principles

- `ggmice` is compatible with the popular R packages `mice` (for imputation) and `ggplot2` (for flexible data visualizations based on the ‘grammar of graphics’ framework).
- `mice` imputation models are substantive model agnostic: flexible, var-by-var definition of imputation model. `ggmice` supports this.
- Grammar of Graphics = layered plots (Wilkinson, 2012) → `ggmice` produces `ggplot` objects: layered, easily adjustable
- Assumptions: imputation model should fit the observed part, imputation model should be congenial → providing tools to see observed and imputation side-by-side/evaluation of the distribution of observed and imputed data (Nguyen et al., 2017)

Limitations

- make explicit what these plots/analyses can and cannot tell us: imputation models should fit the observed part of the data, but no way to determinately state miss mech.
- visualization $\neq$ objective truth $\rightarrow$ good imputation model fit/plausible values is not a guarantee for statistical validity
- visualization is not a replacement of evaluation but support, also check metrics such as FMI
- interpretation remains subjective (e.g. convergence, MAR assumption, etc.)

What does **ggmice** aim to solve?

The **ggmice** package aims to solve the following problems:

- no direct graphical comparisons between incomplete and imputed data; side-by-side presentation of the data before and after imputation; Visualizations with **ggmice** can be made to inspect marginal and joint distributions of incomplete variables side-to-side to compare incomplete and imputed data
- no previously existing methods for visualizing **mice** imputation models; complex imputation models are easier to review through visualization (as opposed to console prints/excel exports of predictor matrix)
- existing plotting tools for **mids** objects not easily editable/publication-ready; Visualizations with **ggmice** can be extended and adapted with tidyverse functions to create publication-ready graphs of imputed data (reflecting the uncertainty due to missingness).
- unified toolbox for the development and evaluation of imputation models

Existing solutions

The **ggmice** package fills a hiatus in the software landscape, with other packages generally supporting either **mice** or **ggplot2**, but not both. Table REF provides an overview of R packages that offer visualization tools for incomplete and imputed data. In short: with **mice** you cannot plot incomplete data distributions, all other packages lack support for **mice**-imputed data, and visualization of **mice** imputation models did not exist at all prior to **ggmice**.

Table 1: Differential functionality of **ggmice** versus other R packages

	ggmice	mice	nanian	VIM
Incomplete data (e.g., <code>data.frame</code> object)				
– missing data patterns	+	+	$\pm$	+
– joint & marginal distributions	+	–	+	+
Imputation model building				

Table 1: Differential functionality of **ggmice** versus other R packages

	ggmice	mice	naniar	VIM
– associations in observed data	+	–	–	–
– associations with missingness indicators	+	+	–	–
– conditional distribution with missingness indicator	+	–	–	–
– imputation models	+	–	–	–
Imputed data (e.g., mids object)				
– trace plot of the imputation algorithm	+	+	–	–
– joint & marginal distributions	+	+	–	–
User utility				
– extendability (i.e., non-deterministic graph types)	+	–	±	–
– editability (i.e., design adaptable for publication)	+	–	+	–
– popularity (i.e., monthly downloads)	–	+	±	±

Note. This is an incomplete overview of the R package’s functionalities. The purpose of this table is to display *differences* in functionality based on the features offered by **ggmice**. I intend to show **ggmice**’s added value compared to existing methodology. Symbols denote good/well supported (+), medium/moderate support (±), and low/not supported (–).

Software development

- Open Source Software development/FAIR → R, GitHub, CRAN, Zenodo, ...
- CRAN downloads
- GitHub stars and issues
- StackOverflow mentions/questions
- feedback at conferences
- citations of the package (Zenodo reference)

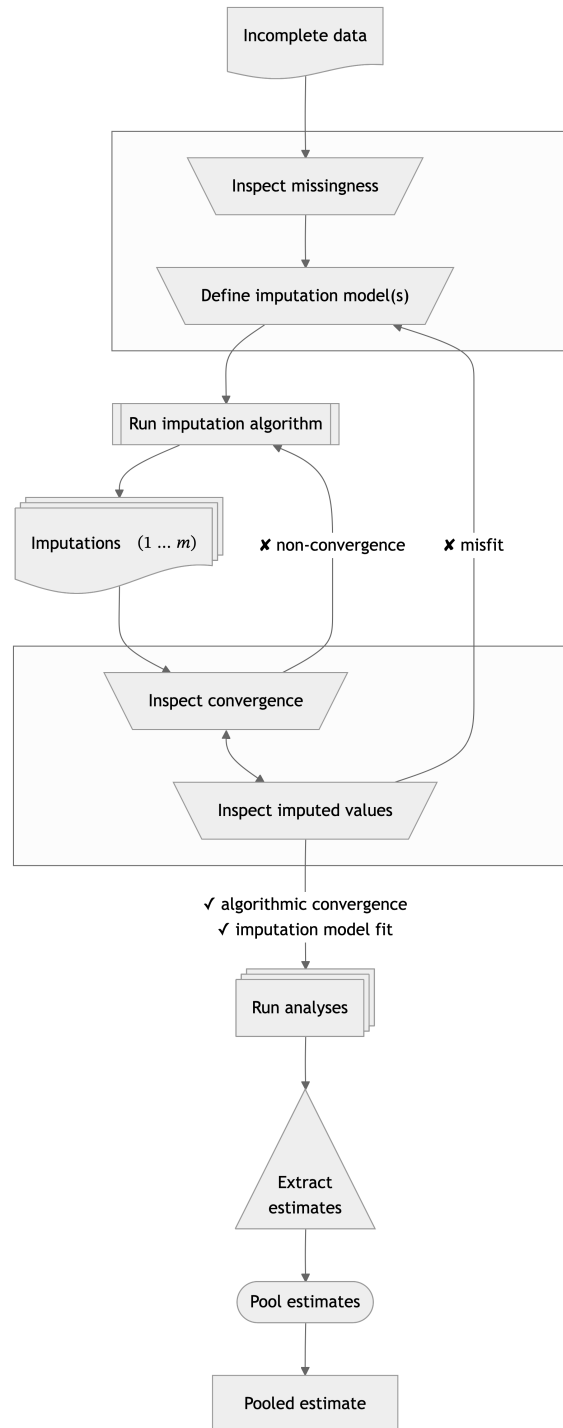
Future research & development

- visualizing **mira** objects
- add posterior predictive check plots & over-imputation
- quantifying uncertainty at pattern/cell level
- tools for sensitivity analyses: inducing MNAR in observed part of the data and evaluating effects on the imputations/inference
- What data-level diagnostics are informative for assessing imputation model adequacy in practice (i.e., when inferential validity cannot be evaluated)?
- pre vs post imputation facilitator functions om te checken of er nieuwe verbanden in zitten (kruistabel)

Building and evaluating imputation models

- Model building: how do we know which functional form to choose?
 - rely on defaults, tested in simulation studies
 - assess the distribution of the incomplete variable (e.g., visual inspection)
 - ...? (maybe ad transformations?)
- Model building: how do we know which imputation model predictors to select?
 - association of incomplete variable with other variables in the data (“include an auxiliary variable if its correlation with the main variable is > 0.5 (in absolute value) [25]” Nguyen et al., 2021)
 - association of missingness indicator with other variables in the data
 - external input (e.g., branching patterns, or expert knowledge to correct for MNAR)
 - Take the analysis model (if available) as starting point for imputation workflow, for congeniality
 - ...?
- Model evaluation: how do we assess imputation model misfit?
 - fit of the model to the observed data
 - non-convergence in the algorithm
 - mismatch between observed and imputed data distributions (e.g., visual inspection) (van Buuren, 2018)
 - ...?

Figure 5: Multiple imputation flowchart



Note. Statistical evaluation is mainly focused on the relation between the incomplete data and pooled estimates. Computational evaluation is in every trapezoid: processes that require human intervention. In the first box, the question is: 'how can we model the missingness?'. Missing in the graph: drawing inference with pooled estimate, and use external info after inspecting the missingness to involve all stakeholders (CRISP-DM steps).

Chapter 1: evaluation

[TODO: link/insert]

Chapter 2: convergence

[TODO: link/insert]

Chapter 3: ggml

[TODO: link/insert]

Discussion

Conclusion

- Computational evaluation of imputation methodology is never finished.
- The main findings in this thesis are...
 - Defining and testing new imputation methods through simulation studies requires careful consideration of the simulation design and evaluation to make imputation methods comparable across studies. We need simulation studies to know imputation methods work. But we cannot be certain they actually *did* work whenever we apply them. The missing part of the data is by definition not available, so we cannot definitively state anything about the imputation model fit to the unobserved part of the data. Moreover, there's untestable assumptions (e.g., MNAR) and lack of clear diagnostic cut offs (e.g., convergence).
 - Non-convergence is hard to diagnose, and typical thresholds to evaluate non-convergence may not be applicable to FCS algorithms: MICE reaches stable output before non-convergence metrics would indicate so. There is a mismatch between theoretical convergence criteria and practical algorithmic behavior. That's why we still have to rely on visual inspection.
 - Visual inspection aids imputers in developing and evaluating imputation models, because formal tests are not available. The R package `ggmice` offers tools for the computational evaluation of the imputations models that were previously unavailable.
- Recommendations for fellow missing data methodologists:
 - design simulation studies structured and reproducible (e.g., report design choices, publish code)
 - talk to your intended audience (i.e., applied researchers), or at least check what they are struggling with (e.g., StackOverflow)
 - ...?
- Recommendations for fellow imputers:
 - think before you code
 - look at the data, plot the data
 - ...?

Limitations

- Computational evaluation is not a substitute for thinking. It may give a false sense of certainty, e.g. against MNAR mechanism (untestable assumption).

- Instead, use computational evaluation as complement to (not a replacement for) substantive knowledge and sensitivity analyses. Use expert insight
- This entire dissertation relies on the R language and centers the `mice` package [TODO rephrase as not a flaw: I focus on a dominant ecosystem, and concepts generalize beyond].
 - While this set-up is very popular, the data science landscape has expanded and is ever evolving
 - Many machine learning and deep learning methods are not (yet) implemented as imputation models in `mice`. These might be able to better approximate the posterior predictive distribution of the missing values than `mice` methods, but research should show that still.
- ~~I haven't tested whether the tools I developed actually improve the imputations of `mice` users~~ This dissertation evaluates tools and diagnostics at the methodological level; assessing their causal impact on user behavior and imputation quality is not feasible to systematically test.
 - GitHub issues and StackOverflow are indication
 - Onmogelijk om evidence-based conclusies te trekken, maar misschien wel evidence-informed?
- The visualizations implemented in `ggmice` are generally better suited for numeric data, as opposed to categorical variables (or even open text field data, which is not supported by `mice` currently). Future research on: associations of cat vars with miss indicators, and convergence parameter other than SD of imputed values.

- TODO: rephrase 'haven't's into future research (positive phrasing: reframe as scope conditions, not shortcomings)
 - maybe future: computational evaluation of variable importance within imputation models

Outlook

- This chapter ends with a future perspective...
- Missing data remains a problem indefinitely. I expect imputation methodology to be developed and applied for the foreseeable future. But the way we deal with data might change.
- With the rise of machine learning and deep learning methods (or “AI”), evaluation will become ever more important. Novel methods are being developed under a prediction framework, not statistical inference framework: ignoring the uncertainty in the estimates,

and thus not incorporating between-imputation variance. This leads to too narrow CIs, and too low p-values. Which, in turn, causes spurious results.

- AI models should propagate uncertainty!
- trust in science requires honest reflection of uncertainty in our analyses
- be open about uncertainty, and limitations of scientific studies
- publish null results (e.g., registered reports or pre-registrations)
- share data and code with other scientists, and the public
- publish open access, educational materials too!
- change the way we evaluate science: this dissertation is an example of the new recognition and rewards vision. it includes software as an actual chapter, not something extra.
- I hope to inspire others to reflect on their own view of what scientific output is. And most of all, I hope to set an example for other PhD candidates to do the same
- imputation is not always the way to go: missingness can tell you about mismatch between data collection method and lived experience → positionality??

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